

Where to investigate first

Purchase-to-pay process intelligence | BPI Challenge 2019

Decision requested: Commission investigations into payment-block handling, invoice clearing waits, and repeated invoice receipts. Assign a separate owner to review nine flagged cases. Validate business context before changing controls or setting improvement targets.

251,734

Purchase-item cases
across the complete log

15,182

Selected-category cases
for conformance review

15,045

Eligible selected cases
for duration analysis

Focus: **3-way match, invoice after GR**. The eligible cohort has **302,275 events**, a **63.10-day median** and **128.30-day P90** observed trace duration. This is the first-to-last recorded span, not necessarily a completed process.

Five investigations, with different evidence types

#	Investigation	Cases / reach	Evidence	Score / band
1	Payment-block prevention and resolution	2,252 14.97%	+35.96 days Median duration difference, with marker minus without	78 P1
2	Invoice receipt to clearing	7,718 51.30%	21.96 days median adjacent-event wait; 65.03-day P90	77 P1
3	Repeated formal invoice receipts	1,221 8.12%	+67.46 days Median duration difference, with marker minus without	75 P1
4	Goods receipt to formal invoice receipt	5,360 35.63%	19.08 days Median adjacent-event wait	70 P2
5	Vendor invoice creation to formal receipt	4,886 32.48%	20.39 days Median adjacent-event wait	70 P2

Reach uses 15,045 eligible cases. Groups overlap; do not add their reach or duration differences. Invoice receipt to clearing accounts for **24.32%** of observed waiting time, which can include agreed payment terms. None of these figures is an estimate of recoverable time.

Protect controls; separate incomplete evidence

9,667 conforming (63.67%) | 5,506 incomplete (36.27%) | 9 review-required (0.06%).
These sum to all 15,182 selected cases. Review the nine individually. Incomplete evidence is not a confirmed violation, and longer blocked-case durations do not justify removing valid controls.

From evidence to investigation

Proposed sequence and owners; no operational intervention or deadline has been agreed.

Workstream	Suggested owner	Evidence needed before a decision
Control review	Finance control owner + data owner	Disposition of all nine flagged cases, based on source event positions and review reasons; explanation of missing milestones in incomplete traces.
Payment blocks	Accounts payable operations	Block reasons; block-to-release and release-to-clearing intervals; separation of necessary controls from preventable input issues.
Clearing waits	Accounts payable + treasury	Reliable due dates and payment terms, separating expected waiting from overdue or avoidable processing time.
Repeated receipts	Invoice processing owner	Classification of partial invoices, corrections, reversals, and possible duplicates before introducing preventive controls.
Pilot decision	P2P manager + data owner	A scoped intervention only if supported: baseline, comparison approach, outcome measures, and control safeguards agreed first.

Measures to establish before a pilot

Use validated avoidable or overdue days, repeat-posting reasons, and exception-resolution time alongside control outcomes. Set numeric targets after confirming definitions and baseline coverage. No savings estimate is supported at this stage.

How the scores were formed

Reach 25% | Time impact 25% | Evidence 25% | Actionability 15% | Confidence 10%.

Components are rated 1-5 and scaled to 100 points. P1 begins at 75; P2 at 65. Weights and business ratings are analyst choices, not probabilities or returns.

Interpretation boundaries

Association is not causation. Seven marker comparisons use two-sided Mann-Whitney U tests, Benjamini-Hochberg adjustment, and rank-biserial effects. Confounding, overlapping markers, related purchase items, and incomplete records remain relevant.

Scope matters. The analysis concerns one category in a historical anonymised log. Directly-follows waits exclude paths with intervening events. Ordering-rule passes do not establish every financial control, and repeated receipts are not automatically duplicate invoices.

Verification and evidence trail

The recorded source-data run passed validation and dashboard checksum gates; **92 automated tests passed**. These support implementation reliability, not causal interpretation or independent validation of every Power BI measure.

Analysis implementation: **5e191f1**. Reviewed dashboard export: **03072ab**.

Detailed findings, methods, and reproduction guide:

github.com/ajin623/processiq/tree/main/docs

Dataset: van Dongen, B. (2019), BPI Challenge 2019, 4TU.Centre for Research Data, CC BY 4.0.

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